

2(a)	<i>Vibrio cholerae</i> ; <i>must be spelled correctly</i>	1
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Question	Answer	Marks
2(b)	any four from: <i>within general response, in correct context</i> 1 ref. to, <u>contamination</u> / <u>contaminated</u> ; A presence of, pathogen / <i>V. cholerae</i> / bacterium <i>in correct context (can look for ora)</i> 2 ref. to breaking the <u>transmission cycle</u> ; <i>providing access to safe drinking water</i> 3 cholera is transmitted by <u>faecal-oral</u> route ; A described e.g. ref. to (contaminated) faeces <u>and</u> ref. to, food washed in contaminated water or drinking (contaminated) water or eating (contaminated) food <i>practise personal hygiene</i> 4 ref. to using method to make sure faeces do not enter drinking water ; e.g. use of, toilets / latrines A do not defaecate in or near, rivers / reservoirs / lakes AW <i>accept</i> urinate I non-scientific terms for, defaecation / urination 5 washing hands ; 6 detail ; e.g. use, soap / antibacterial gel / AW after, urination / defaecation A after going to the toilet before handling foods <i>context is preparation or eating a meal</i> scrub under fingernails cafes, restaurants / AW, provide places for washing hands do not share towels / use disposable towels / AW 7 AVP ; e.g. water, is / should be, chlorinated / boiled / bottled / filtered to remove pathogens / UV-treated / other water, is / should be, kept separate from sewage / piped direct from water treatment plant keep fingernails short safe disposal of vomit	4

Question	Answer			Marks
2(c)(i)	<i>antigen is a standalone mark then move to rows 3,4,5</i> <i>if no marks gained from rows 3,4,5, check for correct answers in these rows going <u>down</u> the column to award 1 mark</i>			4
	row	feature	oral cholera vaccine	passive immunisation for cholera
	1	component causing the desired response	<u>antigen(s)</u> ; I weak / weakened / toxin A toxoid / inactivated toxin R dead antigen A weakened / attenuated / dead, pathogen / <i>V. cholerae</i>	antibody
	2	type of immunity gained	artificial active	artificial passive
	3	stimulates production of memory lymphocytes	<u>yes</u> / no	yes <u>(no)</u> ;
	4	length of time needed to have an effect	shorter same <u>longer</u>	longer same <u>shorter</u> ;
	5	duration of immunity	shorter same <u>longer</u>	longer same <u>shorter</u> ;

Question	Answer	Marks
2(c)(ii)	B-lymphocyte / plasma cell / splenocyte, <u>and</u>, myeloma / tumour, cell ; A B-cell <i>for B-lymphocyte</i> A cancer cell <i>for myeloma cell</i> <i>plus two from:</i> (B-lymphocyte) synthesises (specific / desired) / AW, antibody ; I has the antibody (B-lymphocyte) cannot survive (for long) / dies, in culture ; (myeloma cell) can divide / can carry out mitosis / long-lived ; I rapidly / fast A <u>plasma cell</u> cannot divide R (hybridoma cell) divides uncontrollably	3